Assignment 2

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Assign 1 : Basic Linux Commands		
Please find the commands for the following. (Find appropriate command line options for the commands and write commands solutions into your notebook)		
No.	**Commands**	**To Do**
1.	man	Command to get help about commands
2.	uname	Print System information
3.	pwd	Print current directory
4.	hostname	Print IP address and hostname of the machine
5.	date	Find today's date
6.	cal	Print calendar for current year
7.	ls, dir	See directory structure
8.	cat	Create file
9.	cat >>	Append some text into existing file
10.	cp	Copy a file, Copy a directory
11.	mv	Move the files
12.	mv	Rename a file
13.	tac, rev	Print the file in reverse order, Reverse each line in the file
14.	wc	Count the words, characters and lines in a file
15.	mkdir	Create directory
16.	cd	Change current directory
17.	rm	Remove a file, remove a directory (with -R)
18.	rmdir	remove empty directory
19.	touch	change the modification time , Create Multiple empty files at once
20.	sort	Sort file.
21.	uniq	Avoid repetition in a file.
22.	head	Print the specified number of lines from top in the file.
23.	tail	Print the specified number of lines from bottom in the file.
24.	factor	Find the factors of a number
25.	alias	Create command shortcut
26.	unalias	remove the command shortcut
27.	echo	Print environment variables individually
28.	ps	Print current processes
29.	kill	Kill the process
30.	env	Print environment variables

1.man wc

```
vagrant@ubuntu-jammy:~$ man wc
```

```
WC(1)                          User Commands                          WC(1)

NAME
       wc - print newline, word, and byte counts for each file

SYNOPSIS
       wc [OPTION]... [FILE]...
       wc [OPTION]... --files0-from=F

DESCRIPTION
       Print  newline,  word,  and  byte counts for each FILE, and a total
       line  if  more  than  one  FILE  is  specified.   A   word   is   a
       non-zero-length sequence of characters delimited by white space.

       With no FILE, or when FILE is -, read standard input.

       The  options  below may be used to select which counts are printed,
       always in the following order: newline, word, character, byte, max-
       imum line length.

       -c, --bytes
              print the byte counts

       -m, --chars
              print the character counts

       -l, --lines
              print the newline counts

       --files0-from=F
              read  input from the files specified by NUL-terminated names
              in file F; If F is - then read names from standard input

       -L, --max-line-length
              print the maximum display width

       -w, --words
              print the word counts

       --help display this help and exit

       --version
              output version information and exit
 Manual page wc(1) line 1 (press h for help or q to quit)
```

2.uname

```
vagrant@ubuntu-jammy:~$ uname
Linux
```

3.pwd

```
vagrant@ubuntu-jammy:~$ pwd
/home/vagrant
```

4.hostname

```
vagrant@ubuntu-jammy:~$ hostname
ubuntu-jammy
```

5.date

```
vagrant@ubuntu-jammy:~$ date
Tue Mar 24 15:45:11 UTC 2026
```

6.cal

```
vagrant@ubuntu-jammy:~$ cal
     March 2026
Su Mo Tu We Th Fr Sa
 1  2  3  4  5  6  7
 8  9 10 11 12 13 14
15 16 17 18 19 20 21
22 23 24 25 26 27 28
29 30 31
```

7.ls,dir

```
vagrant@ubuntu-jammy:~$ ls
3.txt  sunbeam
vagrant@ubuntu-jammy:~$ dir
3.txt  sunbeam
```

8.cat

```
vagrant@ubuntu-jammy:~$ cat 3.txt
Hello!
```

9.cat>>

```
vagrant@ubuntu-jammy:~$ cat>>3.txt
Hello!
```

10.cp

```
vagrant@ubuntu-jammy:~$ cp 3.txt sunbeam/
```

11.mv(rename)

```
vagrant@ubuntu-jammy:~$ mv 3.txt 33.txt
```

12.mv(move)

```
vagrant@ubuntu-jammy:~$ mv 33.txt sunbeam/
```

13.tac,rev

tac-

```
vagrant@ubuntu-jammy:~/sunbeam$ tac 3.txt
10
9
8
7
6
5
4
3
2
1
```

rev-

```
vagrant@ubuntu-jammy:~/sunbeam$ rev 33.txt
!olleH
```

14.wc-

```
vagrant@ubuntu-jammy:~/sunbeam$ wc file.txt
wc -l file.txt
wc -w file.txt
wc -c file.txt
 4  4 58 file.txt
4 file.txt
4 file.txt
58 file.txt
```

15. mkdir

```
vagrant@ubuntu-jammy:~/sunbeam$ mkdir files.txt
vagrant@ubuntu-jammy:~/sunbeam$ ls
18oct  3.txt  33.txt  data.txt  file.txt  files.txt  grade.txt  one
```

16.cd

```
vagrant@ubuntu-jammy:~/sunbeam$ cd 18oct/
vagrant@ubuntu-jammy:~/sunbeam/18oct$
```

17.rm

```
vagrant@ubuntu-jammy:~/sunbeam$ rm file.txt
vagrant@ubuntu-jammy:~/sunbeam$ ls
18oct  3.txt  33.txt  data.txt  files.txt  grade.txt  one
```

18.rmdir

```
vagrant@ubuntu-jammy:~/sunbeam$ ls
18oct  3.txt  33.txt  data.txt  grade.txt  one
```

19.touch

```
vagrant@ubuntu-jammy:~/sunbeam$ touch file.txt
vagrant@ubuntu-jammy:~/sunbeam$ ls
18oct  3.txt  33.txt  data.txt  file.txt  grade.txt  one
```

20.sort

```
vagrant@ubuntu-jammy:~/sunbeam$ sort file.txt
sort -r file.txt
sort -n file.txt
sort -u file.txt
```

21.uniq

```
vagrant@ubuntu-jammy:~/sunbeam$ uniq file.txt
uniq -c file.txt
uniq -d file.txt
uniq -u file.txt
```

22.head

```
vagrant@ubuntu-jammy:~/sunbeam$ head -3 file.txt
10 + 20
bc
cat > input.txt
```

23.tail

```
vagrant@ubuntu-jammy:~/sunbeam$ tail -3 file.txt
egrep "good|cake" taste.txt
grep "good|cake" taste.txt
grep "is" -R .
```

24.factor

```
vagrant@ubuntu-jammy:~/sunbeam$ factor 10
10: 2 5
```

25.alias,26.unalias

```
vagrant@ubuntu-jammy:~/sunbeam$ alias sus='date'
vagrant@ubuntu-jammy:~/sunbeam$ date
Tue Mar 24 16:06:43 UTC 2026
vagrant@ubuntu-jammy:~/sunbeam$ unalias sus
```

27.echo

```
vagrant@ubuntu-jammy:~/sunbeam$ echo "Hello"
echo $HOME
echo -n "Hello"
echo -e "Hello\nWorld"
Hello
/home/vagrant
HelloHello
World
```

28.ps

```
vagrant@ubuntu-jammy:~/sunbeam$ ps
    PID TTY          TIME CMD
   2404 pts/0    00:00:01 bash
   3186 pts/0    00:00:00 ps
```

29.kill

```
vagrant@ubuntu-jammy:~/sunbeam$ kill 1
-bash: kill: (1) - Operation not permitted
```

30.env

```
vagrant@ubuntu-jammy:~/sunbeam$ env
env | grep PATH
env VAR=value command
SHELL=/bin/bash
PWD=/home/vagrant/sunbeam
LOGNAME=vagrant
XDG_SESSION_TYPE=tty
MOTD_SHOWN=pam
HOME=/home/vagrant
LANG=C.UTF-8
LS_COLORS=rs=0:di=01;34:ln=01;36:mh=00:pi=40;33:so=01;35:do=01;35:bd=40;33;01:cd=40;33;01:or=40;31;01:mi=00:su=37;41:sg=30;43:ca=30;41:tw=30;42:ow=34;42:st=
37;44:ex=01;32:*.tar=01;31:*.tgz=01;31:*.arc=01;31:*.arj=01;31:*.taz=01;31:*.lha=01;31:*.lz4=01;31:*.lzh=01;31:*.lzma=01;31:*.tlz=01;31:*.txz=01;31:*.tzo=01
;31:*.t7z=01;31:*.zip=01;31:*.z=01;31:*.dz=01;31:*.gz=01;31:*.lrz=01;31:*.lz=01;31:*.lzo=01;31:*.xz=01;31:*.zst=01;31:*.tzst=01;31:*.bz2=01;31:*.bz=01;31:*.
tbz=01;31:*.tbz2=01;31:*.tz=01;31:*.deb=01;31:*.rpm=01;31:*.jar=01;31:*.war=01;31:*.ear=01;31:*.sar=01;31:*.rar=01;31:*.alz=01;31:*.ace=01;31:*.zoo=01;31:*.
cpio=01;31:*.7z=01;31:*.rz=01;31:*.cab=01;31:*.wim=01;31:*.swm=01;31:*.dwm=01;31:*.esd=01;31:*.jpg=01;35:*.jpeg=01;35:*.mjpg=01;35:*.mjpeg=01;35:*.gif=01;35
:*.bmp=01;35:*.pbm=01;35:*.pgm=01;35:*.ppm=01;35:*.tga=01;35:*.xbm=01;35:*.xpm=01;35:*.tif=01;35:*.tiff=01;35:*.png=01;35:*.svg=01;35:*.svgz=01;35:*.mng=01;
35:*.pcx=01;35:*.mov=01;35:*.mpg=01;35:*.mpeg=01;35:*.m2v=01;35:*.mkv=01;35:*.webm=01;35:*.webp=01;35:*.ogm=01;35:*.mp4=01;35:*.m4v=01;35:*.mp4v=01;35:*.vob
=01;35:*.qt=01;35:*.nuv=01;35:*.wmv=01;35:*.asf=01;35:*.rm=01;35:*.rmvb=01;35:*.flc=01;35:*.avi=01;35:*.fli=01;35:*.flv=01;35:*.gl=01;35:*.dl=01;35:*.xcf=01
;35:*.xwd=01;35:*.yuv=01;35:*.cgm=01;35:*.emf=01;35:*.ogv=01;35:*.ogx=01;35:*.aac=00;36:*.au=00;36:*.flac=00;36:*.m4a=00;36:*.mid=00;36:*.midi=00;36:*.mka=0
0;36:*.mp3=00;36:*.mpc=00;36:*.ogg=00;36:*.ra=00;36:*.wav=00;36:*.oga=00;36:*.opus=00;36:*.spx=00;36:*.xspf=00;36:
SSH_CONNECTION=10.0.2.2 65462 10.0.2.15 22
LESSCLOSE=/usr/bin/lesspipe %s %s
XDG_SESSION_CLASS=user
TERM=xterm-256color
LESSOPEN=| /usr/bin/lesspipe %s
USER=vagrant
SHLVL=1
XDG_SESSION_ID=7
XDG_RUNTIME_DIR=/run/user/1000
SSH_CLIENT=10.0.2.2 65462 22
XDG_DATA_DIRS=/usr/local/share:/usr/share:/var/lib/snapd/desktop
PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin
DBUS_SESSION_BUS_ADDRESS=unix:path=/run/user/1000/bus
SSH_TTY=/dev/pts/0
_=/usr/bin/env
OLDPWD=/home/vagrant/sunbeam/18oct
PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin
env: ‘command’: No such file or directory
vagrant@ubuntu-jammy:~/sunbeam$
```